Adidas US Sales Analysis (Python)

Exploratory data analysis of Adidas US sales using Python: data cleaning, feature engineering, and static and interactive visualizations to understand sales and profit across regions, products, sales methods, and time.

Business Questions

- Which regions generate the most sales?
- Which products contribute the most to operating profit?
- Which days of the week are the most profitable?
- How do sales and profit change over time and by sales method (In-store, Online, Outlet)?

Dataset

- **Source:** Adidas US Sales dataset (add the link here)
- **Size:** 9,648 rows and 12 columns (9,641 rows after cleaning)
- **Period:** January 2020 to December 2021
- **Columns:** Retailer, Retailer ID, Invoice Date, Region, State, City, Product, Price per Unit, Units Sold, Total Sales, Operating Profit, Sales Method

Tools

Python, Pandas, NumPy, Matplotlib, Seaborn, Plotly Express, Jupyter Notebook

Workflow

1. Data Cleaning

- Checked for duplicate rows (none found).
- Found missing values: 7 in `Region` and 2 in `Price per Unit`.
- Filled the missing prices using the average price of the same state and region.
- Removed the 7 rows with a missing region (9,641 rows remained).
- Converted `Invoice Date` from Excel serial numbers to real dates.

2. Feature Engineering

- Created a `Day` column (day of the week) to analyze profit by weekday.
- Calculated each product's share of total operating profit.

3. Analysis

- Total sales by region
- Profit share by product
- Average operating profit by day of the week

4. Visualization

- **Matplotlib and Seaborn:** profit and sales over time, sales trend by sales method, number of transactions by sales method, average sales by method, and a joint plot of sales over time.
- **Plotly (interactive):** profit by product and sales method, sales share by method (pie chart), and sales by state and region (sunburst).

Key Insights

1. **The West region leads in sales** with about $26.9M, roughly 30% of the total, followed by the Northeast ($18.6M) and the Southeast ($16.3M). The South ($14.4M) and the Midwest ($13.6M) are the lowest.
2. **Men's Street Footwear is the top profit driver** with about 25% of total operating profit, followed by Women's Apparel (20.7%) and Men's Athletic Footwear (15.6%).
3. **Profit is well diversified:** no product category exceeds a quarter of total profit, and the six main products each contribute between 11.7% and 25%.
4. **Thursday and Friday are the most profitable days:** the average profit per transaction is about $4,000 on Thursday and $3,800 on Friday, compared with about $3,000 on Saturday, the lowest day.

Recommendations

- Keep investing in the **West region**, and study what drives the weaker results in the Midwest and South.
- Prioritize inventory and promotions for **Men's Street Footwear** and **Women's Apparel**.
- Schedule campaigns and stock replenishment around **Thursday and Friday**, when profit per transaction is highest.